Overview of T-SQL in SQL Server

T-SQL (Transact-SQL) is the **procedural extension of ANSI SQL** used in **Microsoft SQL Server**.
It allows developers and database administrators to **query data, modify data, manage database objects, and implement business logic directly inside the database**.

While **ANSI SQL** provides the standard commands for interacting with relational databases, **T-SQL adds programming capabilities** such as:

- Variables
- Conditional statements
- Loops
- Error handling
- Stored procedures
- Transactions
- Functions

Because of these features, **T-SQL acts almost like a programming language inside SQL Server**.

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1. What is T-SQL?

Definition

T-SQL is Microsoft's proprietary extension of SQL used to interact with SQL Server databases.

It is used to:

- Create databases and tables
- Insert, update, delete data
- Retrieve data
- Create stored procedures and functions
- Control transactions
- Handle errors

2. Why T-SQL is Important

In database-driven applications such as **banking systems, e-commerce systems, and ERP systems**, T-SQL is used to:

1. Manage large volumes of data
2. Implement business rules
3. Improve performance using stored procedures
4. Maintain data integrity
5. Control transactions

Example:
Bank transaction logic may run inside T-SQL procedures.

Example:

```
BEGIN TRANSACTION;

UPDATE Accounts
SET Balance = Balance - 1000
WHERE AccountID = 1;

COMMIT;
```

If an error occurs:

```
ROLLBACK;
```

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5. Programming Features of T-SQL

Unlike standard SQL, T-SQL supports **programming logic**.

Variables

Variables store temporary values.

Example:

```
DECLARE @TotalBalance INT;

SET @TotalBalance = 5000;

SELECT @TotalBalance;
```

Conditional Statements

T-SQL supports **IF…ELSE conditions**.

Example:

```
DECLARE @Balance INT = 1000;

IF @Balance > 500
   PRINT 'Sufficient Balance';
ELSE
   PRINT 'Low Balance';
```

Loops

Used to repeat execution.

Example:

```
DECLARE @Counter INT = 1;
```

```
WHILE @Counter <= 5
BEGIN
   PRINT @Counter;
   SET @Counter = @Counter + 1;
END
```

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6. Database Objects Created Using T-SQL

T-SQL is used to create many database objects.

Object	Description
Views	Virtual tables
Stored Procedures	Precompiled SQL logic
Functions	Return values
Triggers	Automatic execution on events
Indexes	Improve query performance

Example:

```
CREATE VIEW CustomerView
AS
SELECT Name, City
FROM Customers;
```

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7. T-SQL Stored Procedures

Stored procedures contain **precompiled SQL code stored in the database**.

Example:

```
CREATE PROCEDURE GetCustomers
```

```
AS
BEGIN
   SELECT * FROM Customers;
END
```

Execute:

```
EXEC GetCustomers;
```

Benefits:

- Faster execution
- Security
- Reusability
- Reduced network traffic

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8. T-SQL Functions

Functions return values.

Types:

1. Scalar Functions
2. Table-Valued Functions

Example:

```
CREATE FUNCTION GetTax(@Amount INT)
RETURNS INT
AS
BEGIN
   RETURN @Amount * 10 / 100;
END
```

Usage:

```
SELECT dbo.GetTax(1000);
```

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9. T-SQL Exception Handling

T-SQL provides **TRY...CATCH blocks**.

Example:

```
BEGIN TRY
   SELECT 10 / 0;
END TRY
BEGIN CATCH
   PRINT 'Error occurred';
END CATCH
```

This prevents query failure.

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10. T-SQL Transactions

Transactions ensure **ACID properties**:

Property	**Meaning**
Atomicity	All or nothing
Consistency	Data remains valid
Isolation	Transactions independent
Durability	Changes are permanent

Example:

```
BEGIN TRANSACTION;

UPDATE Accounts
SET Balance = Balance - 500
WHERE AccountID = 1;
```

```
UPDATE Accounts
SET Balance = Balance + 500
WHERE AccountID = 2;

COMMIT;
```

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11. Advantages of T-SQL

1. Powerful procedural programming
2. High performance
3. Built-in error handling
4. Supports transactions
5. Works with large enterprise applications
6. Secure database operations

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13. Where T-SQL is Used

T-SQL is used in:

- Banking systems
- E-commerce applications
- ERP systems
- Data warehouses
- Reporting systems
- Web applications (ASP.NET, APIs).

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14. Difference Between SQL and T-SQL

Feature	SQL	T-SQL
Standard	ANSI standard	Microsoft extension
Programming features	Limited	Advanced
Variables	No	Yes
Error handling	No	TRY CATCH
Loops	No	WHILE
Stored procedures	Limited	Full support

✅ **Simple Summary**

T-SQL is the **programming language used to interact with SQL Server databases**.
It extends SQL with **procedural programming, transactions, error handling, and advanced database operations**.